Lesson plan 2026 / 27

Day & Date	Week	Period/s	Lesson Title	Subject	Chapter/Unit	Grade/ section
Week 7 · day per timetable	7	1 period (60 min)	Properties of Logarithms	Algebra II	Topic 6 — Exponential and Logarithmic Functions · Lesson 6-5	Grade 11 — 11B

Tools To Be Used	Resources
Quizizz (diagnostic) · Pear Deck (live practice feedback) · Scientific calculator (LOG and LN keys) · Canva (final product) · Whiteboards	Lesson presentation (14 slides) · "Choose Your Route" differentiation sheet · "Red Sea Tourism Growth" project task (A3) · "Give it a go" classwork · squared paper

Competencies	Technology Embedded	Connection to student's real life
1- Expanding and condensing logarithmic expressions using the product, quotient and power properties. 2- Rewriting and evaluating logarithms with the change of base formula, exactly and with a calculator.	1- Quizizz live diagnostic with instant gap analysis. 2- Calculator LOG and LN keys: the same value from either, rounded only at the last step.	1- A visitor model for a Red Sea destination: in which year does it pass a target? 2- Planners who need the year a target is reached, and the revenue in that year (SAR).

Values	Soft Skills	Hard Skills
1- Precision — rounding only at the last step. 2- Long-term planning — using logarithms to plan for national targets.	1- Collaboration in assigned group roles. 2- Explaining in words why a property works.	1- Expanding, condensing and evaluating logarithms. 2- Finding the term number of a geometric sequence with a logarithm.

Lesson Integral Parts

Objectives / skills	Teacher Strategies & Student Activities	Assessment	Time
1- Apply properties of logarithms to rewrite logarithmic expressions and evaluate logarithms. 2- Use the change of base formula to rewrite and evaluate logarithmic expressions. *Essential question: How can the properties of logarithms, including the change of base formula, be applied to simplify and evaluate logarithmic expressions?* *Vocabulary: change of base formula ; logarithmic properties.*	FIKR Phase 1 — Diagnose. 3-question Quizizz: $\log_2 32$, $2^3 \cdot 2^4$ as one power, $\log_{10} 0.01$. Results read as a gap map; students routed to the re-teach table, straight to the properties, or the Investigate prompt. FIKR Phase 2 — Targeted Instruction. Whole class: products and quotients; powers; the change of base formula; logarithms and geometric sequences. Quick check on whiteboards. FIKR Phase 3 — Practice. Two identical "Give it a go" problems worked together. Students then choose a route and may switch at any time. Live feedback via Pear Deck; 2-minute micro-conferences. FIKR Phase 4 — Production. The Red Sea visitor model: terms, the year a target is passed, the revenue in SAR. AI used as critic only. Extended in the A3 project task. FIKR Phase 5 — Proof of Learning. "Time to Check" — condense, evaluate with change of base, and a given-value logarithm. No notes, no partner, no AI. FIKR Phase 6 — Smart Production. Students refine their reasoning into a final product including one worked non-example, and publish to the LMS portfolio.	Diagnostic quiz — gap map, not a grade. Quick check for understanding on whiteboards (80% threshold). "Give it a go" questions — self-marked, teacher circulates. Project task sheet against its "Done when…" criteria. "Time to Check" — Mastery Gate, scored live. PASS → Enrichment & Challenge. NOT YET → Targeted Learning Clinic next lesson. Final product graded on reasoning and clarity, not correctness alone.	6 min 18 min 12 min 12 min 6 min 6 min **Total 60 min**

Lesson Closure:	Two minutes of peer show-and-tell on the final products, then the reflection question: "Which is harder for you — choosing the right property, or choosing the right base for the change of base formula?"
Reflection for Improvement:	
Homework:	Finish the final product and upload to the LMS portfolio (any of the three formats). Practice set on Pear Deck — 12 questions, choose your route. Watch the flipped video for the next lesson, Topic 6 · Lesson 6-6 — Exponential and Logarithmic Equations and Inequalities. Clinic group: bring your Time to Check paper.
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